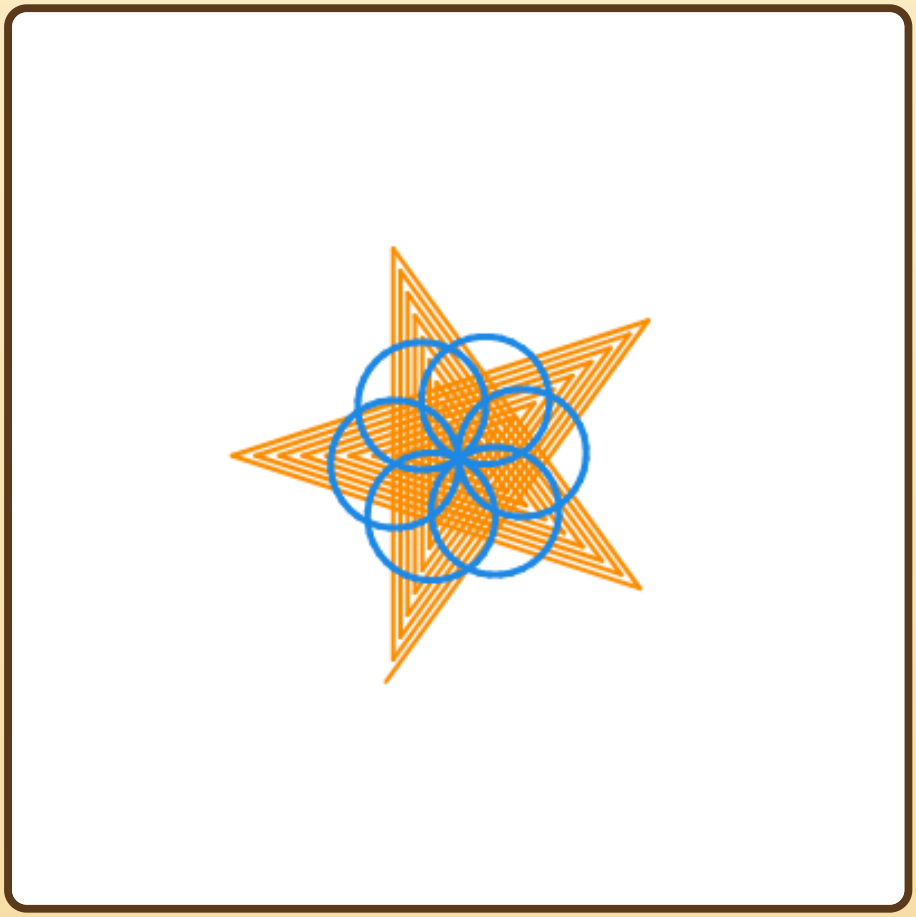
For kids and teachers

Learn Programming with Turtle Drawing

From blocks to Python — 18 drawing challenges with a turtle



National Institute of Technology, Maizuru College — Yasuhito Inoue

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How to use this book

Use this book together with the “Block Drawing” app on a computer or tablet. You will learn how to think like a programmer by building programs with blocks and with Python.

1. Read the **problem** and look at the picture.
2. Think about the order: trace it with your finger or write it in your notebook.
3. Build the program in the app and press “▶ Run”. If it doesn’t work, find what was different and fix it.
4. Stuck? Look at the 💡 **Hint**. When you are done, turn the page and check the 📖 **Answer** and the ✨ **Key point**.

In Drawing, you are correct when your picture matches the light gray “model”.

There is more than one right answer. If it works, your program is correct, even if it is different from the book. Mistakes are hints — try again and again!

Apps in this book

App	Tasks	Grade
Block Drawing	9	Grade 3+ (angles, polygons)
Python Drawing	9	Grade 5 – junior high

PART 1

Part 1 The drawing book for kids

Chapter 1

Draw by giving commands to a turtle

In this app you give commands to a turtle. The turtle holds a pen, and it leaves a line wherever it moves.

Command	What the turtle does
move forward ○ steps	Moves ○ steps the way it is facing (draws a line)
turn right ○ degrees	Turns right ○ degrees on the spot (does not move)
repeat ○ times	Repeats the commands inside ○ times
count with i from ○ to ○ by ○	Repeats while changing the number i (a variable)

The secret of the turn angle

When the turtle turns all the way around, it turns **360 degrees**. To draw a square, the turtle turns 4 times and faces the same way again, so each turn is 360 ÷ 4 = **90 degrees**. For a hexagon, 360 ÷ 6 = 60 degrees.

> The turn angle is not the corner inside the shape. It is how much the turtle **changes the direction it walks**. Try turning your own body to check.

When it doesn’t work

- Compare with the model and find where it starts to go wrong.
- Check that (repeat times) × (turn angle) is 360 (or 720).
- Finding and fixing mistakes is called **debugging**.

Chapter 2

🐢 Block Drawing

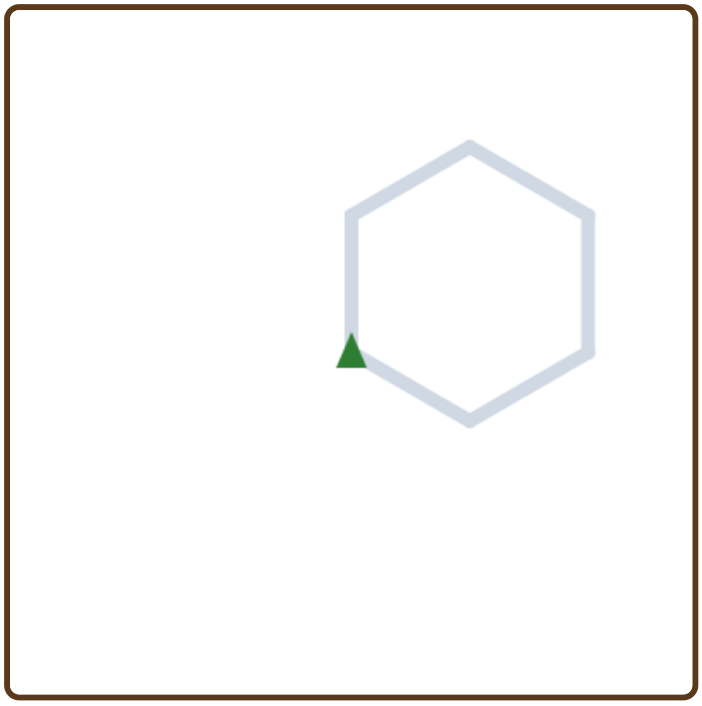
Snap blocks together to draw hexagons, stars and circles, then spirals and snowflakes. You will go from **loops → nested loops → variables → breaking a task into parts**.

Commands in this chapter

move forward / move back / turn right, left ◯ degrees / repeat ◯ times / count with i from ◯ to ◯ by ◯ / math (+ − × ÷) / pen color

① Hexagon ★

Loops Angles



Model (draw over the light lines)

Problem

Draw a hexagon with sides of 80 steps.

🎯 **Goal**：Draw a regular polygon with a loop

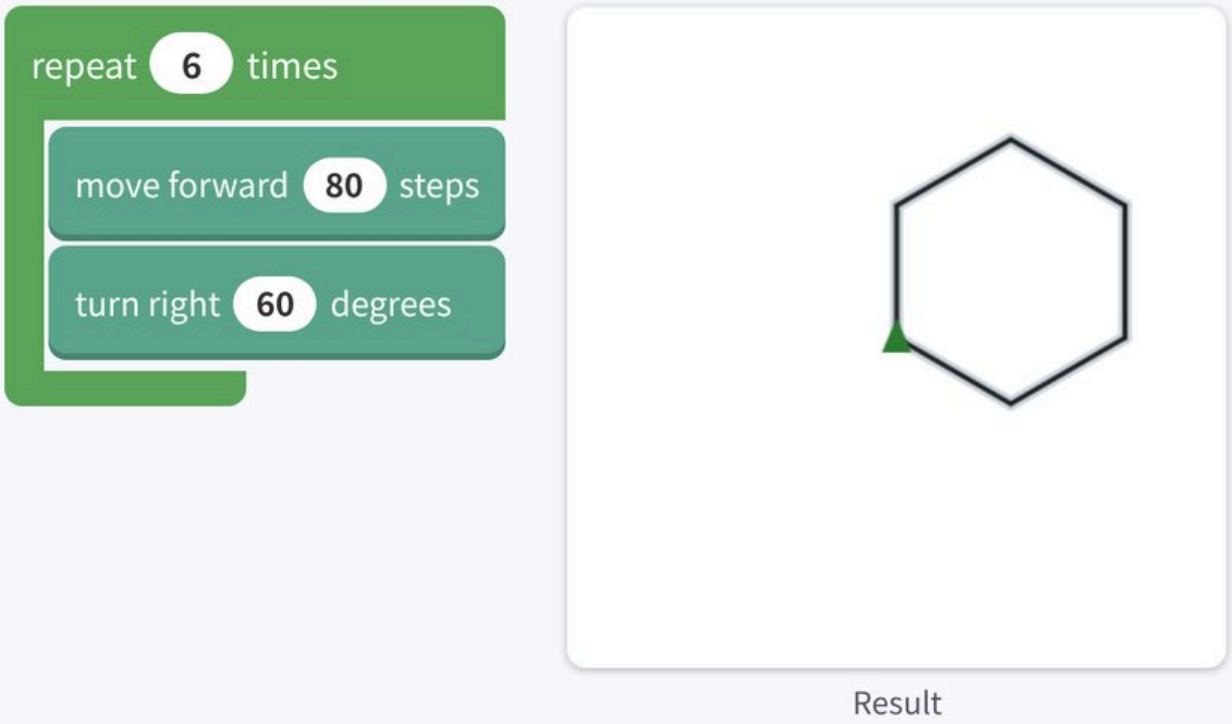
💡 **Hint (look if you get stuck)**

Repeat 6 times. How many degrees to turn? 360 ÷ 6 = ?

✏ **Write your own idea**

The answer is on the next page →

📖 Answer ① Hexagon ★



✨ Key point

Repeat 6 times and turn 60° each time: 360 ÷ 6 = 60.

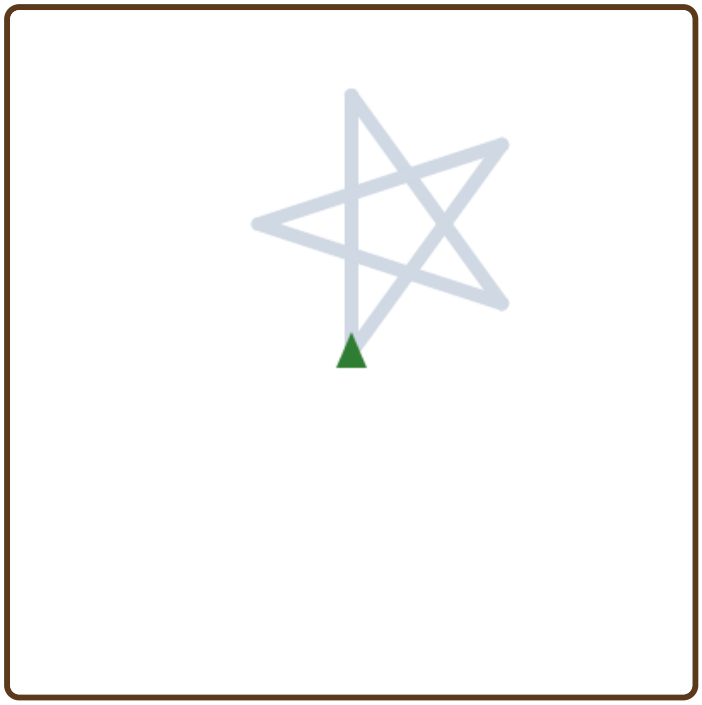
⚠ Common mistake

Turning 120° (the inside angle). The turtle turns the outside angle.

🎬 Video: `videos/en/draw-01.mp4`

② Star ★★

Loops Angles



Model (draw over the light lines)

Problem

Draw a star with lines of 150 steps.

🎯 **Goal**：Draw a shape that turns around twice

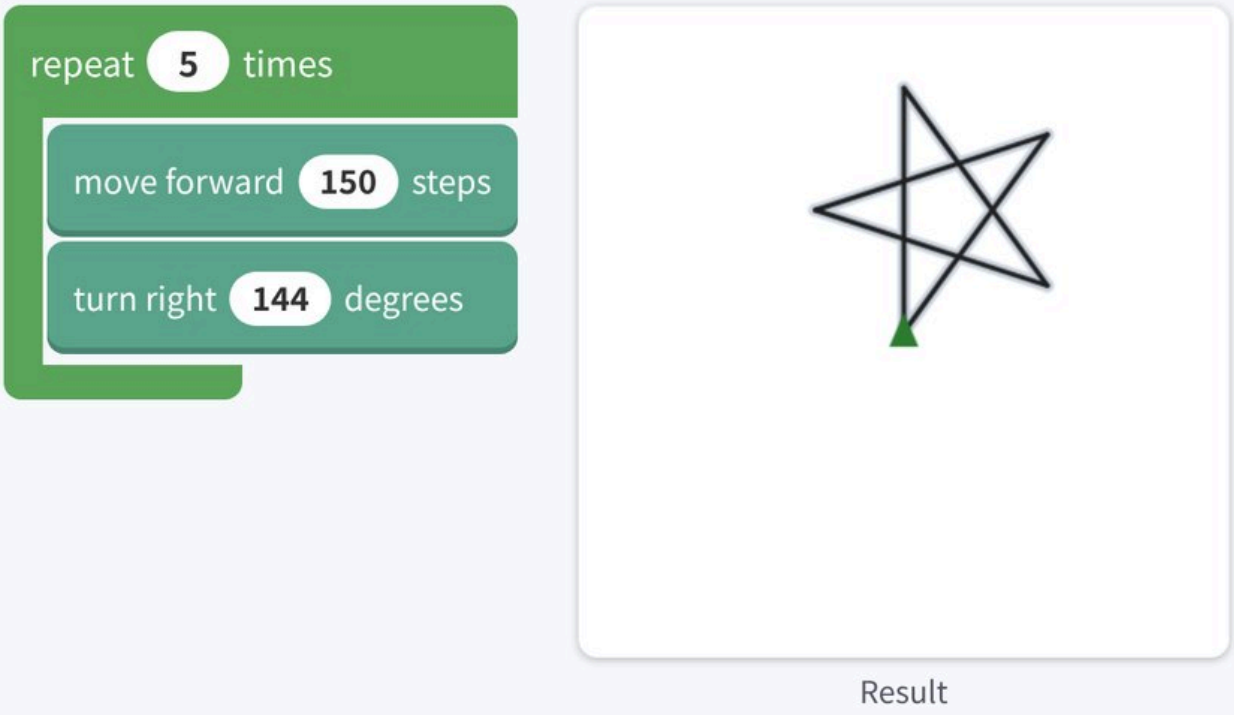
💡 **Hint (look if you get stuck)**

After 5 repeats the turtle turns around twice (720 degrees). 720 ÷ 5 = ?

✏ **Write your own idea**

The answer is on the next page →

📖 Answer ② Star ★★



✨ Key point

It turns 720° (twice around) in 5 repeats, so each turn is 720 ÷ 5 = 144°.

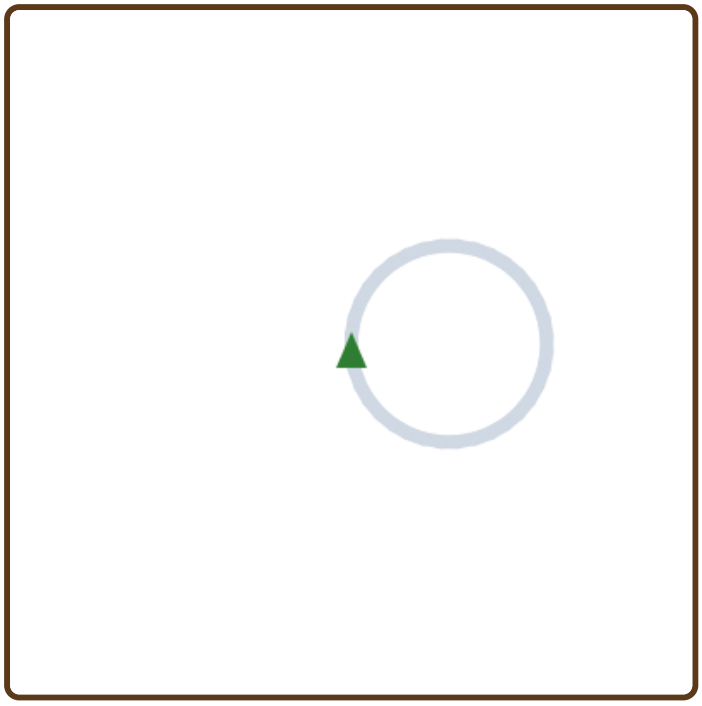
⚠ Common mistake

Using 72° (360 ÷ 5) draws a pentagon, not a star.

🎬 Video: `videos/en/draw-02.mp4`

③ Circle ★★

Loops Angles



Model (draw over the light lines)

Problem

Draw a circle by repeating: move 10 steps, then turn a little.

🎯 **Goal**：Make a circle from short lines

💡 **Hint (look if you get stuck)**

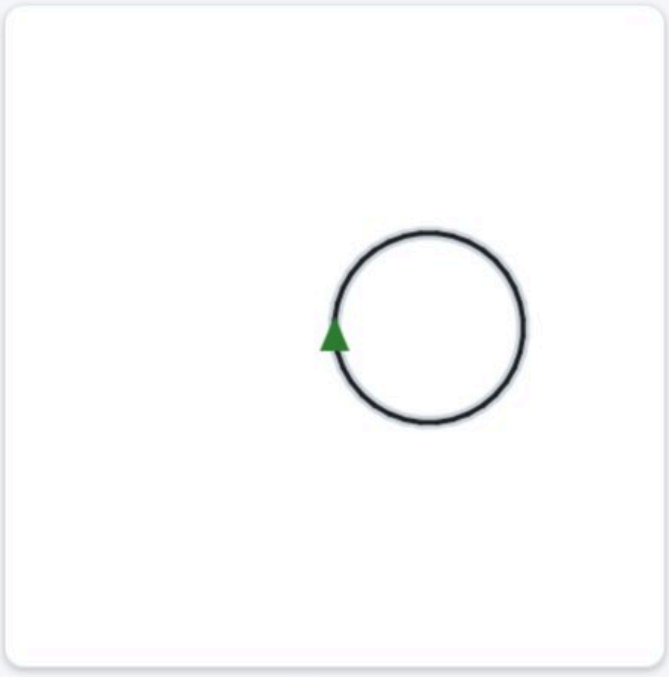
One full turn is 360 degrees. If you turn 10 degrees each time, how many repeats do you need?

✏ **Write your own idea**

The answer is on the next page →

📖 Answer ③ Circle ★★





Result

✨ **Key point**

Move 10 and turn 10°, 36 times. Many small turns look like a circle.

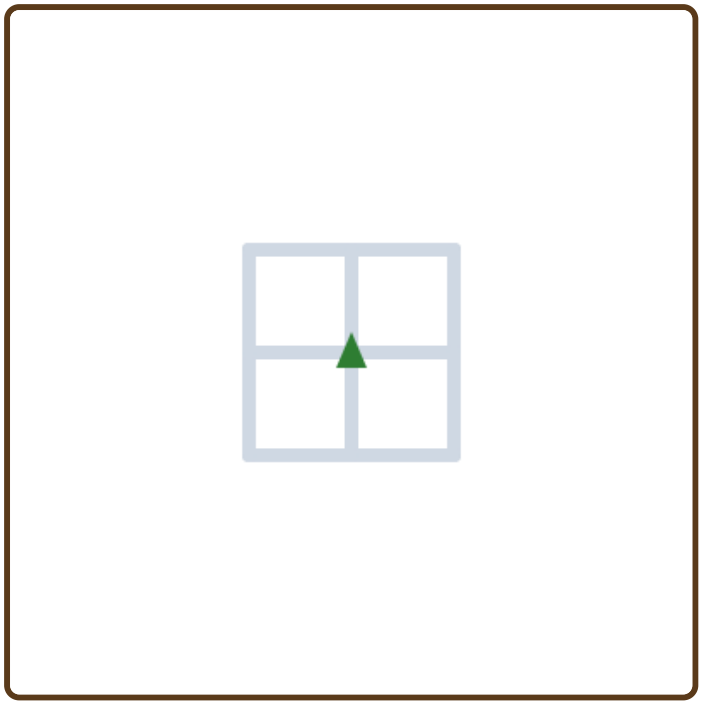
⚠ **Common mistake**

Too few repeats and the circle stops partway (360 ÷ 10 = 36).

🎬 Video: `videos/en/draw-03.mp4`

④ Window ★★★

Nesting



Model (draw over the light lines)

Problem

Draw 4 squares (sides of 60 steps) next to each other to make a window.

🎯 **Goal**：Put a loop inside a loop

💡 **Hint (look if you get stuck)**

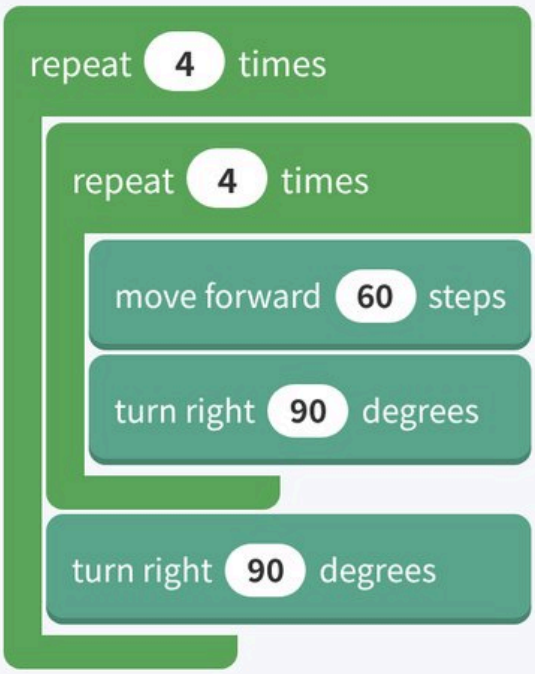
Repeat "draw a square" 4 times. After each square, turn right 90 degrees.

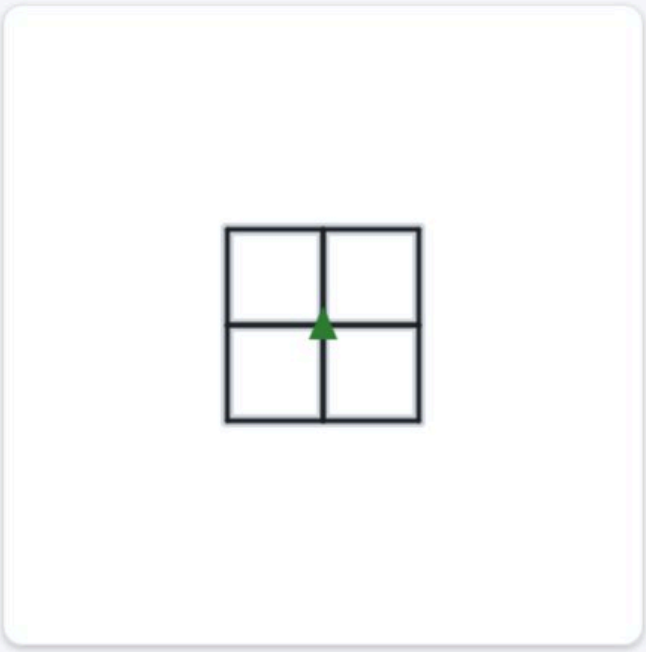
Put a repeat inside a repeat!

✏ **Write your own idea**

The answer is on the next page →

📖 Answer ④ Window ★★★





Result

✨ Key point

Repeat the square-drawing loop 4 more times.

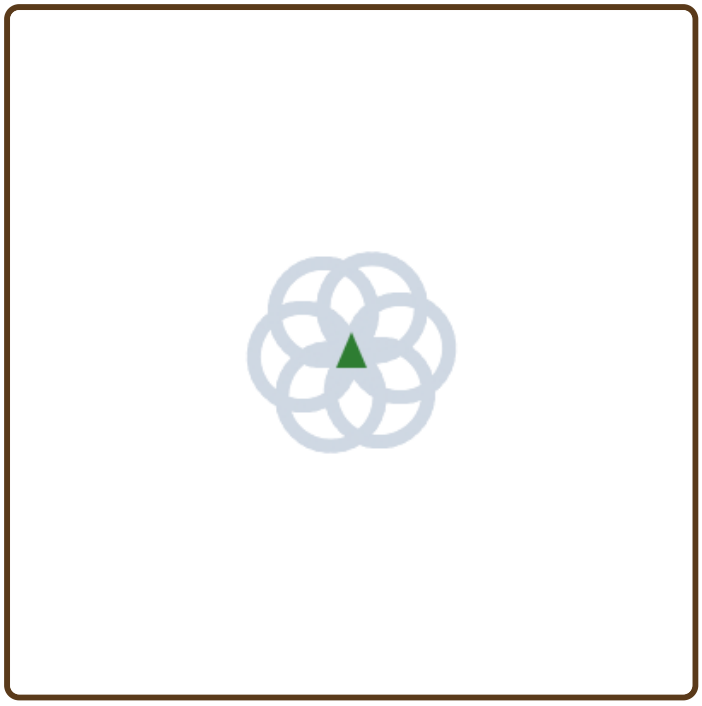
⚠ Common mistake

Forget “turn right 90” after each square, and the squares overlap into one.

🎬 Video: `videos/en/draw-04.mp4`

⑤ Flower ★★★

Nesting Angles



Model (draw over the light lines)

Problem

Draw 6 circles, turning between them, to make a flower. (Circle: 5 steps and 10 degrees, 36 times)

🎯 **Goal**：Nesting + rotating

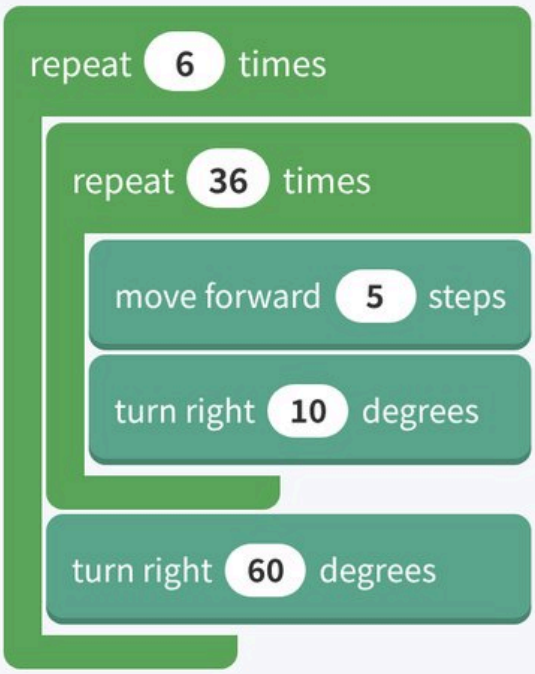
💡 **Hint (look if you get stuck)**

After each circle, turn 360 ÷ 6 = 60 degrees. Repeat this 6 times.

✏ **Write your own idea**

The answer is on the next page →

📖 Answer ⑤ Flower ★★★





Result

✨ **Key point**

After each circle, turn 60°. Do this 6 times.

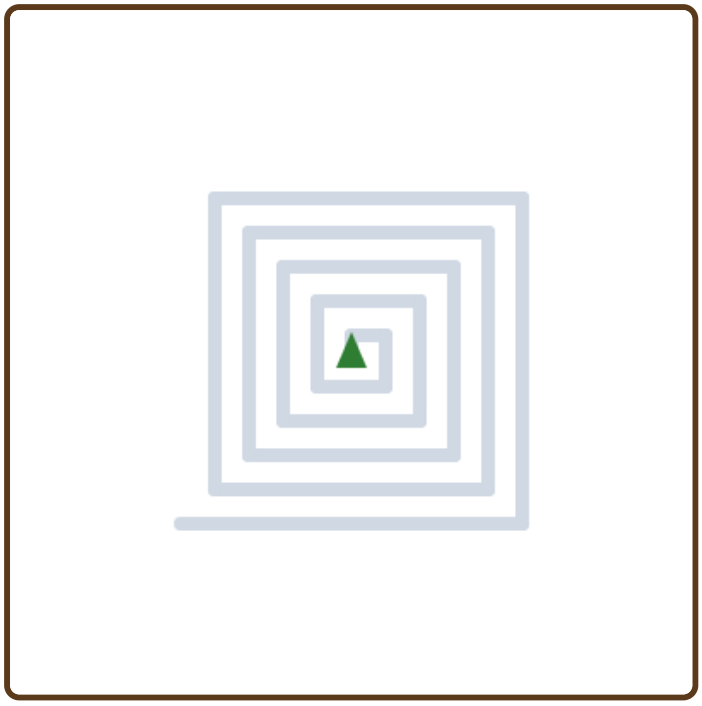
⚠ **Common mistake**

If the turn is not 60°, the petals are uneven.

🎬 Video: `videos/en/draw-05.mp4`

⑥ Square spiral ★★★★

Variables Loops



Model (draw over the light lines)

Problem

Move 10, 20, 30 … up to 200 steps, turning right 90 degrees each time.

🎯 **Goal**：Use a variable as a length

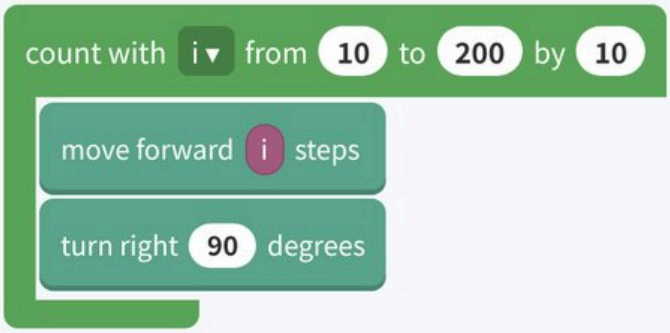
💡 **Hint (look if you get stuck)**

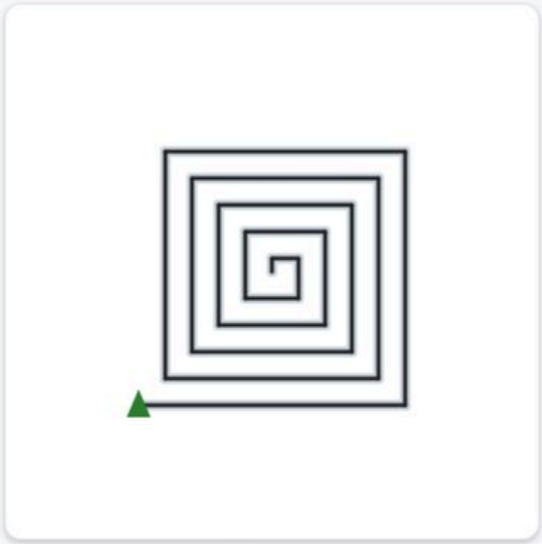
Use the loop "count with i from 10 to 200 by 10". Put the variable i into the "move" block.

✏ **Write your own idea**

The answer is on the next page →

📖 Answer ⑥ Square spiral ★★★★





Result

✨ Key point

i changes to 10, 20, 30..., so the line gets longer and longer.

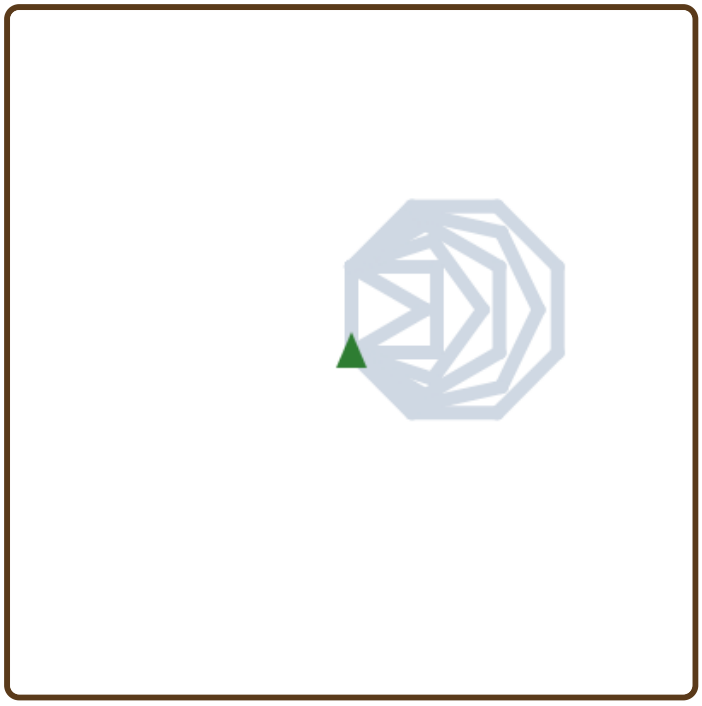
⚠ Common mistake

With a fixed number in “move”, the length never changes. Put the variable i in.

🎬 Video: `videos/en/draw-06.mp4`

⑦ Polygons in a row ★★★★

Variables | Math | Nesting



Model (draw over the light lines)

Problem

From the same spot, draw a triangle, square, pentagon … up to an octagon, each with sides of 50 steps.

🎯 **Goal**：Variable + math

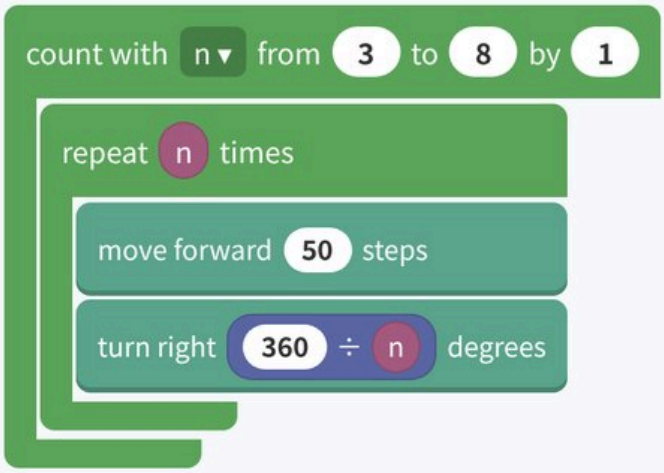
💡 **Hint (look if you get stuck)**

Change n from 3 to 8: "repeat n times: move 50 steps, turn right 360 ÷ n degrees". Use the math block!

✏ **Write your own idea**

The answer is on the next page →

📖 Answer ⑦ Polygons in a row ★★★★





Result

✨ Key point

An n-sided shape turns 360 ÷ n each time. Change n from 3 to 8 to draw 6 shapes.

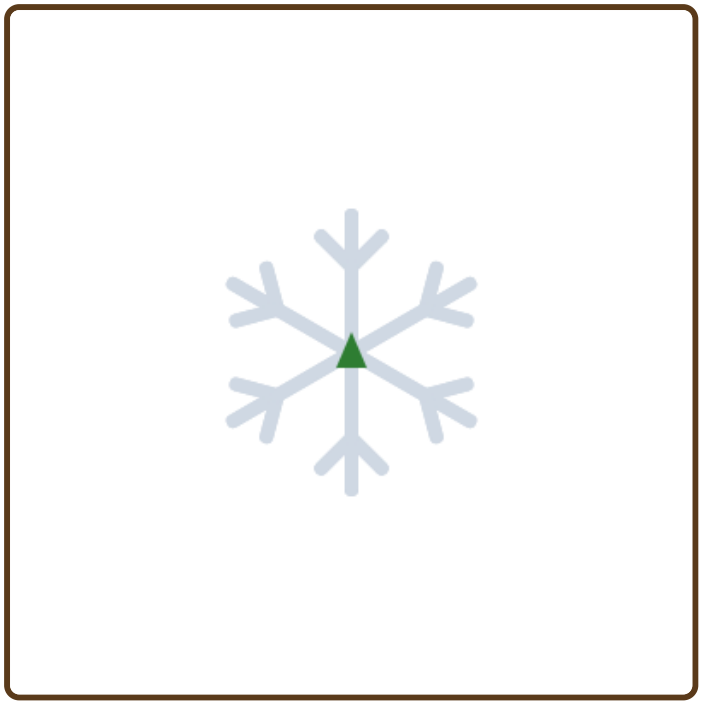
⚠ Common mistake

With a fixed turn (like 60), every shape is the same.

🎬 Video: `videos/en/draw-07.mp4`

⑧ Snowflake ★★★★★

Decomposition Loops



Model (draw over the light lines)

Problem

Draw one branch, come back to where you started, and turn 60 degrees. Repeat this 6 times.

🎯 **Goal**：Break the task into steps

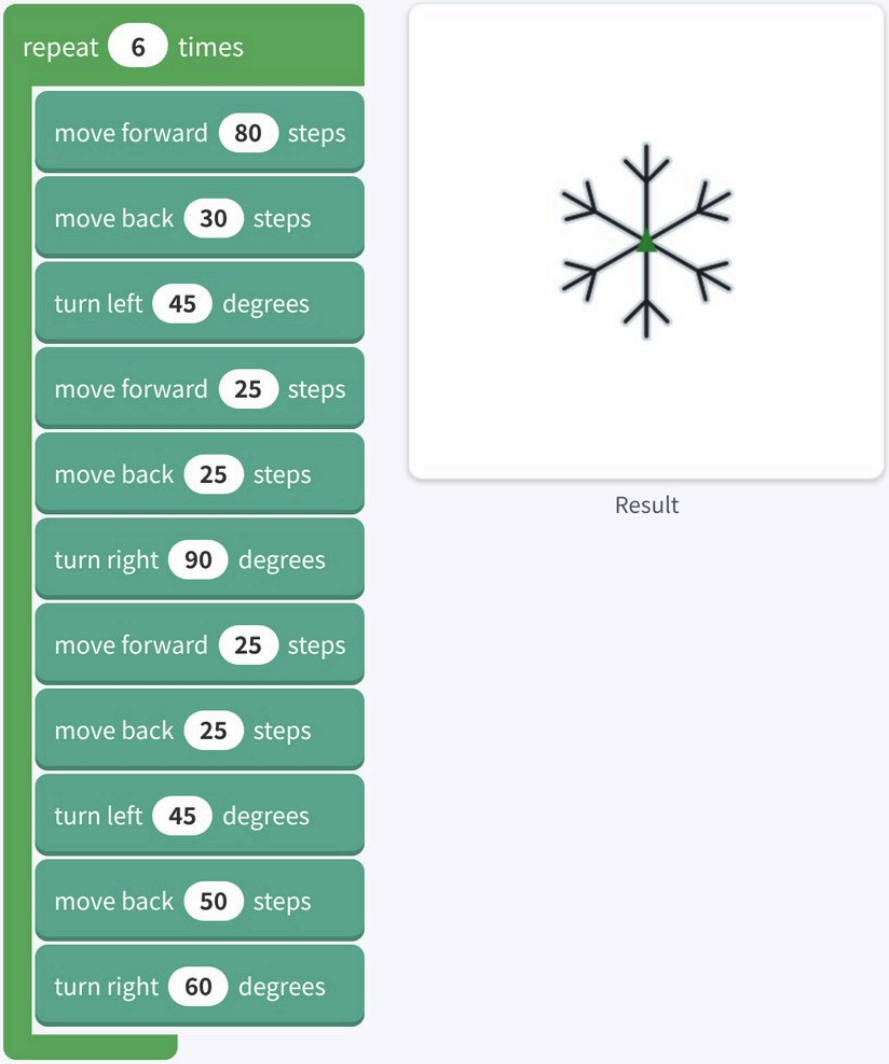
💡 **Hint (look if you get stuck)**

Branch: forward 80 → back 30 → left 45 → forward 25 → back 25 → right 90 → forward 25 → back 25 → left 45 → back 50.

✏ **Write your own idea**

The answer is on the next page →

📖 Answer ⑧ Snowflake ★★★★★



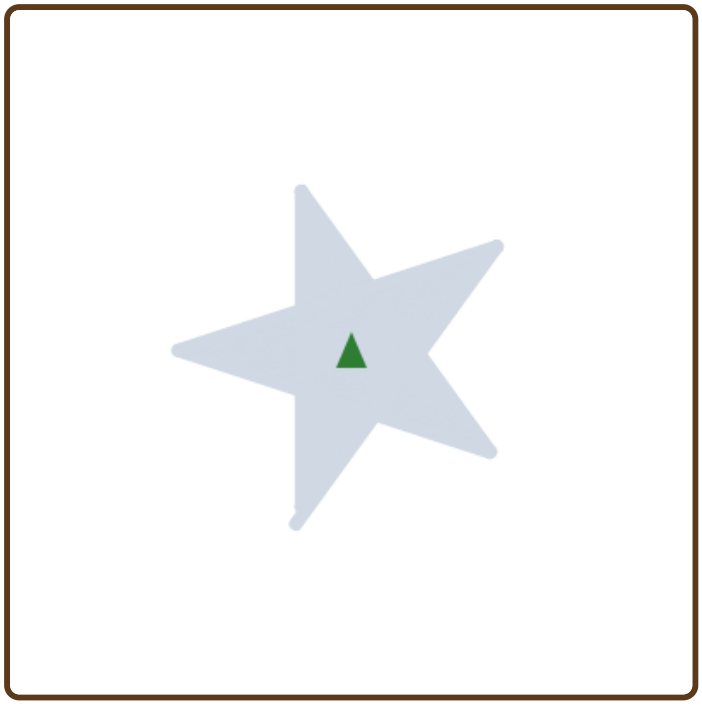
> ✨ **Key point**
>
> The trick: after each branch, put the position and the direction back.

> ⚠ **Common mistake**
>
> Forget the final “move back 50”, and the branches drift.

🎬 Video: `videos/en/draw-08.mp4`

⑨ Star spiral ★★★★★

Variables Math



Model (draw over the light lines)

Problem

Move 4, 8, 12 … steps (4 more each time, 50 times), turning right 144 degrees.

🎯 **Goal**：Variable × math + turning

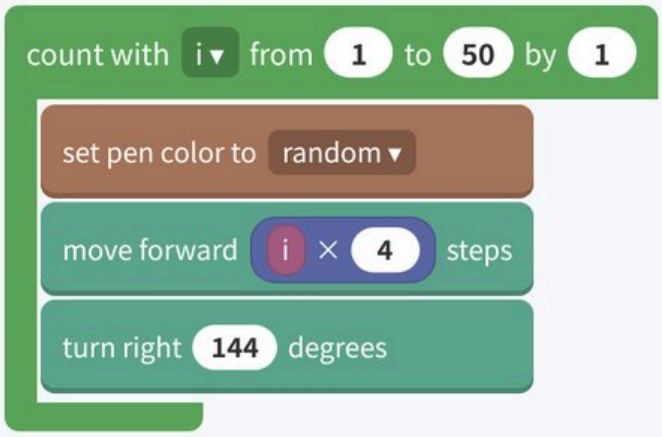
💡 **Hint (look if you get stuck)**

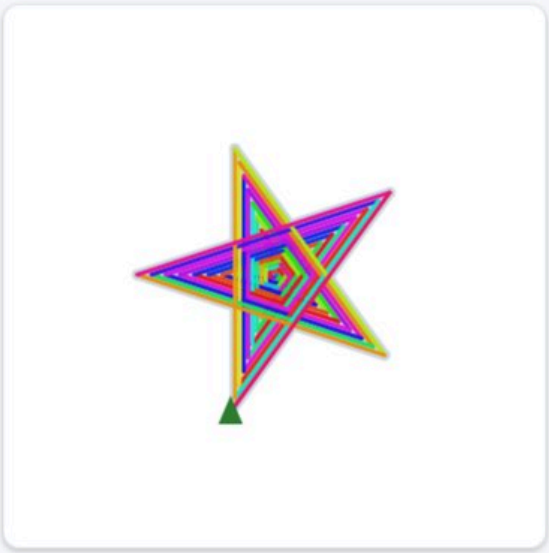
Count i from 1 to 50 by 1. Move i × 4 steps. Try a "random" pen color!

✏ **Write your own idea**

The answer is on the next page →

📖 Answer ⑨ Star spiral ★★★★★





Result

✨ Key point

The length i × 4 grows, and turning 144° each time makes a star spiral.

⚠ Common mistake

Without the variable i in the multiply block, the length does not grow.

🎬 Video: `videos/en/draw-09.mp4`

Column

Corners and turn angles

A regular polygon is “move the same length → turn the same angle”, repeated. The turn angle is always **360 ÷ number of corners**. A star turns around twice, so 720 ÷ 5 = 144 degrees. As you add more corners, the shape gets closer to a circle. With a program, you can check this right away just by changing a number.

Chapter 3

🐍 Drawing in Python

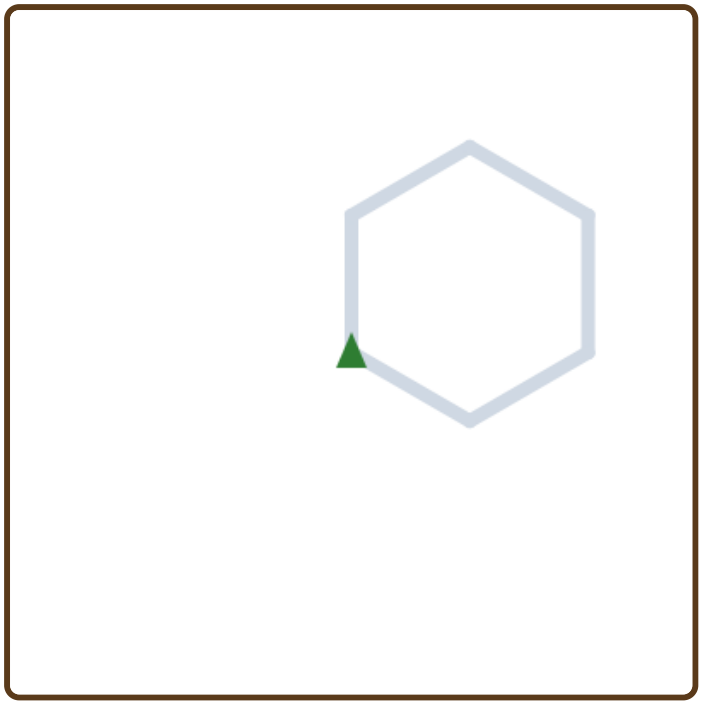
Draw the same challenges in **Python**. “move forward 80 steps” is `forward(80)`, “turn right 60 degrees” is `right(60)`, and “repeat 6 times” is `for i in range(6):`.

Commands in this chapter

forward(80) / right(60) / left(45) / for i in range(6): / for i in range(10, 201, 10): / 360 / n

① Hexagon ★

Loops Angles Python syntax (colon & indent)



Model (draw over the light lines)

Same shape as block challenge 1

Problem

forward(80) moves 80 steps and right(60) turns 60 degrees. Repeat them 6 times to draw a hexagon.

🎯 **Goal** : Draw a regular polygon with for

💡 **Hint (look if you get stuck)**

Put forward(80) and right(60) inside for i in range(6):. 360 ÷ 6 = 60

✏ **Write your own idea**

The answer is on the next page →

📖 Answer ① Hexagon ★

```
for i in range(6):
    forward(80)
    right(60)
```

✨ Key point

Write numbers inside forward() and right(). The 2 lines inside for repeat 6 times.

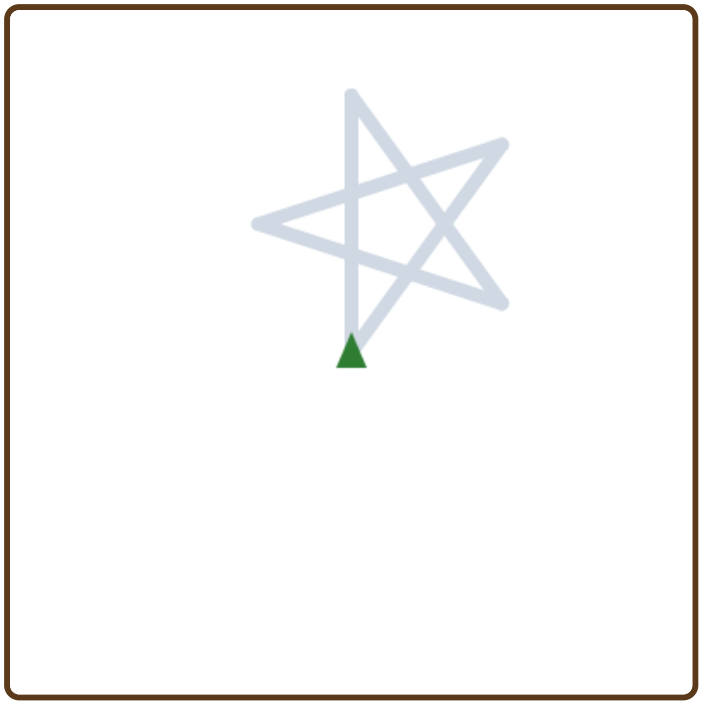
⚠ Common mistake

If right(60) is not indented, it turns only once and draws a straight line.

🎬 Video: `videos/en/pydraw-01.mp4`

② Star ★★

Loops | Angles | Python syntax (colon & indent)



Model (draw over the light lines)

Same shape as block challenge 2

Problem

Draw a star with lines of 150 steps: repeat forward(150) and right(144) 5 times.

🎯 **Goal**：Think about turning around twice

💡 **Hint (look if you get stuck)**

The turtle turns around twice (720 degrees). 720 ÷ 5 = 144

✏ **Write your own idea**

The answer is on the next page →

📖 Answer ② Star ★★

```
for i in range(5):
    forward(150)
    right(144)
```

✨ Key point

5 turns make 720 degrees (twice around), so each turn is 720 ÷ 5 = 144.

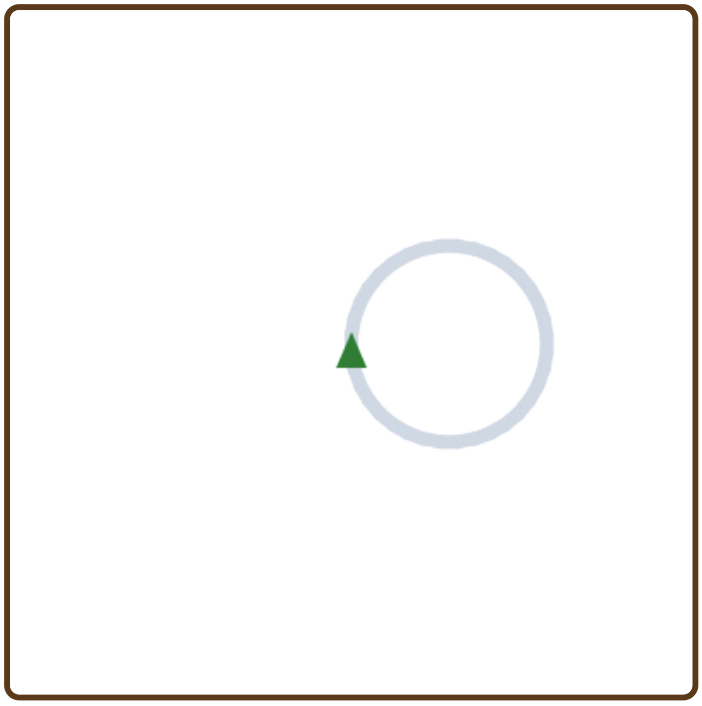
⚠ Common mistake

right(72) (360 ÷ 5) makes a pentagon, not a star.

🎬 Video: `videos/en/pydraw-02.mp4`

③ Circle ★★

Loops | Angles | Python syntax (colon & indent)



Model (draw over the light lines)

Same shape as block challenge 3

Problem

Draw a circle by repeating forward(10) and right(10).

🎯 **Goal**：Make a circle from short lines

💡 **Hint (look if you get stuck)**

One full turn is 360 degrees. 360 ÷ 10 = 36, so use range(36).

✏ **Write your own idea**

The answer is on the next page →

📖 Answer ③ Circle ★★

```
for i in range(36):
    forward(10)
    right(10)
```

✨ Key point

36 turns of 10 degrees make 360. Many short lines look like a circle.

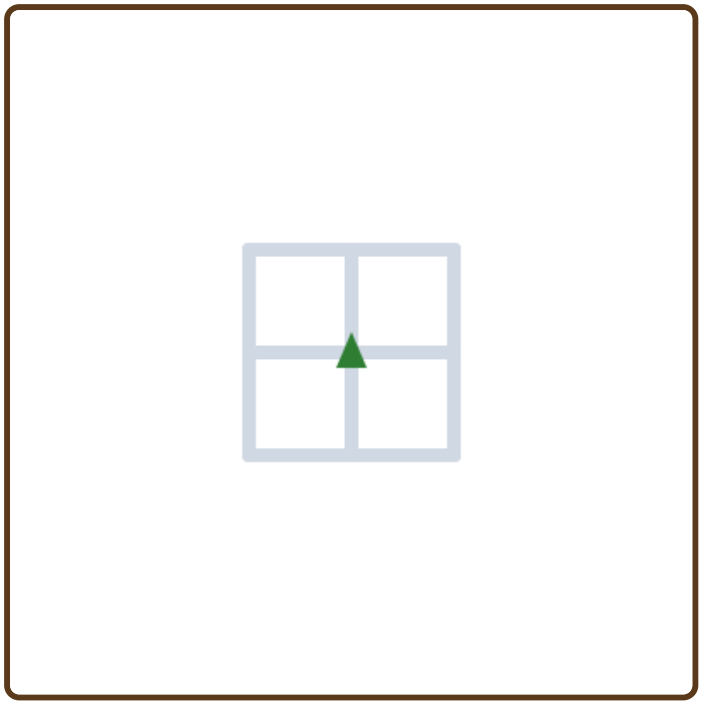
⚠ Common mistake

range(35) leaves a small gap. Check 360 ÷ 10 = 36.

🎬 Video: `videos/en/pydraw-03.mp4`

④ Window ★★★

Loops | Nesting | Python syntax (colon & indent)



Model (draw over the light lines)

Same shape as block challenge 4

Problem

Draw 4 squares (sides of 60 steps) to make a window. Put a for loop inside a for loop.

🎯 **Goal**：Put a for inside a for

💡 **Hint (look if you get stuck)**

Outer loop 4 times: the inner for k in range(4): draws a square, then right(90).

✏ **Write your own idea**

The answer is on the next page →

📖 Answer ④ Window ★★★

```
for i in range(4):
    for k in range(4):
        forward(60)
        right(90)
    right(90)
```

✨ Key point

The inner for is indented 8 spaces. After a square, right(90) inside the outer for.

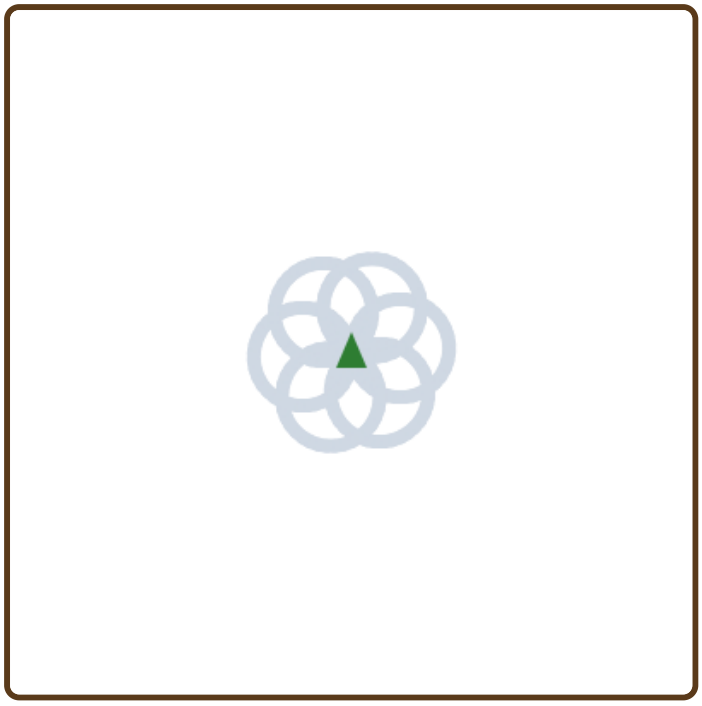
⚠ Common mistake

If the last right(90) is at the inner for’s depth, the squares overlap and look like one.

🎬 Video: `videos/en/pydraw-04.mp4`

⑤ Flower ★★★

Loops | Nesting | Angles | Python syntax (colon & indent)



Model (draw over the light lines)

Same shape as block challenge 5

Problem

Draw 6 circles (forward(5) and right(10), 36 times), turning 60 degrees between them, to make a flower.

🎯 **Goal**：Nesting + turning around

💡 **Hint (look if you get stuck)**

for i in range(6): → for k in range(36): forward(5), right(10) → right(60)

✏ **Write your own idea**

The answer is on the next page →

📖 Answer ⑤ Flower ★★★

```
for i in range(6):
    for k in range(36):
        forward(5)
        right(10)
    right(60)
```

✨ **Key point**

One circle is the inner for. The outer for turns 360 ÷ 6 = 60 degrees, 6 times.

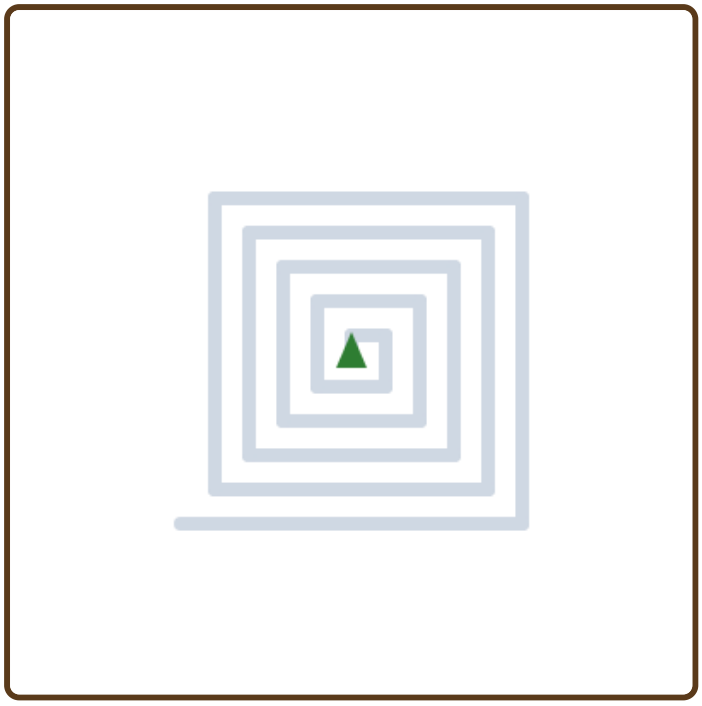
⚠ **Common mistake**

If the turn is not 60, the petals are uneven.

🎬 Video: `videos/en/pydraw-05.mp4`

⑥ Square spiral ★★★★

Variables | Loops | Python syntax (colon & indent)



Model (draw over the light lines)

Same shape as block challenge 6

Problem

Use for i in range(10, 201, 10): so i becomes 10, 20, 30 ... 200, and draw a spiral with forward(i) and right(90).

🎯 **Goal**：Use the variable i as the length

💡 **Hint (look if you get stuck)**

range(start, end, step). The end number is not included, so use 201.

✏ **Write your own idea**

The answer is on the next page →

📖 Answer ⑥ Square spiral ★★★★

```
for i in range(10, 201, 10):
    forward(i)
    right(90)
```

✨ Key point

range(10, 201, 10) gives 10, 20, ... 200. The end is not included, so use 201. forward(i) gets longer each time.

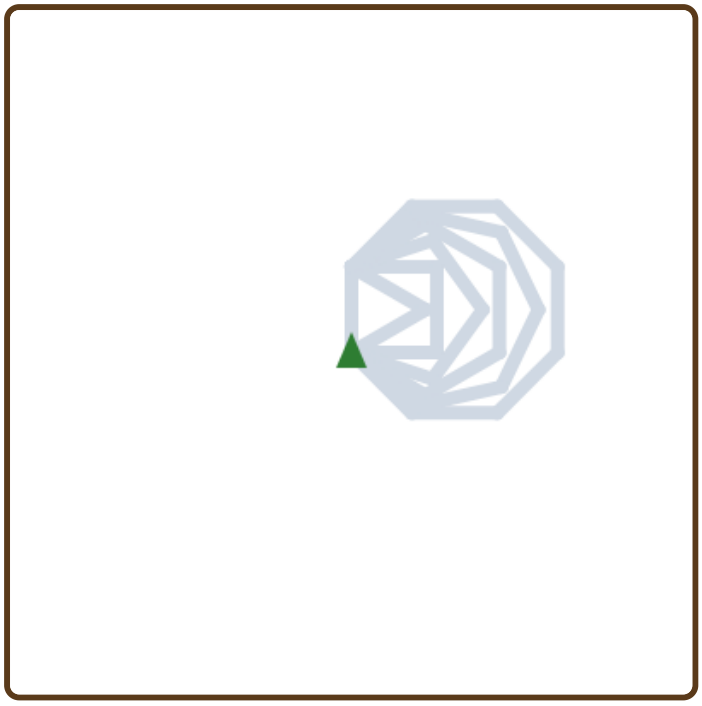
⚠ Common mistake

range(10, 200, 10) stops at 190.

🎬 Video: `videos/en/pydraw-06.mp4`

⑦ Polygons in a row ★★★★

Variables | Math | Nesting | Python syntax (colon & indent)



Model (draw over the light lines)

Same shape as block challenge 7

Problem

Use for n in range(3, 9): to change n from 3 to 8, and draw an n-sided shape (sides of 50 steps) from the same spot.

🎯 **Goal** : Variables and division

💡 **Hint (look if you get stuck)**

Inside, for i in range(n): forward(50) and right(360 / n). / means divide.

✏ **Write your own idea**

The answer is on the next page →

📖 Answer ⑦ Polygons in a row ★★★★

```
for n in range(3, 9):
    for i in range(n):
        forward(50)
        right(360 / n)
```

✨ **Key point**

An n-sided shape turns 360 / n. range(3, 9) gives n = 3 to 8.

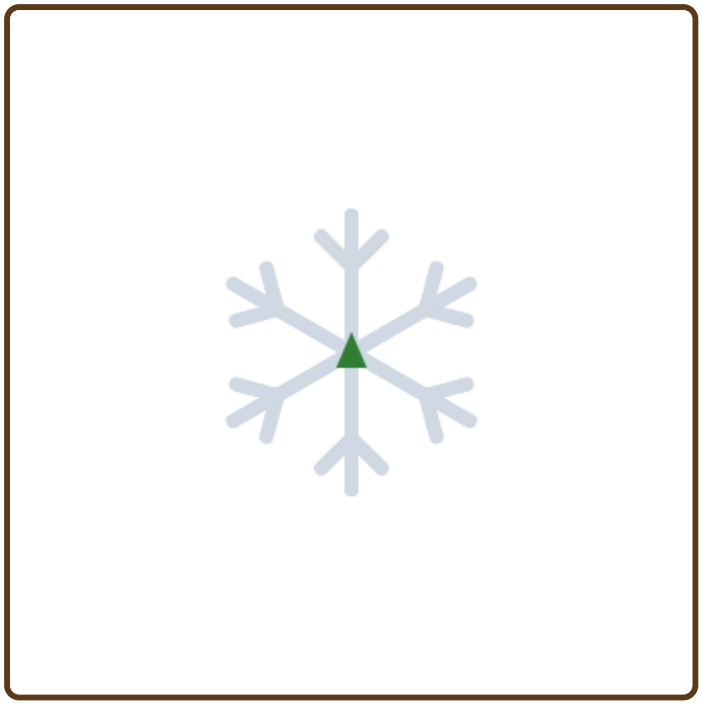
⚠ **Common mistake**

360 // n (floor division) has no block. Use / for division.

🎬 Video: `videos/en/pydraw-07.mp4`

⑧ Snowflake ★★★★★

Sequence | Decomposition | Loops | Python syntax (colon & indent)



Model (draw over the light lines)

Same shape as block challenge 8

Problem

Draw one branch, come back, and right(60). Repeat this 6 times to draw a snowflake.

🎯 **Goal**：Break the steps into parts

💡 **Hint (look if you get stuck)**

Branch: forward(80) → back(30) → left(45) → forward(25) → back(25) → right(90) → forward(25) → back(25) → left(45) → back(50)

✏ **Write your own idea**

The answer is on the next page →

📖 Answer ⑧ Snowflake ★★★★★

```
for i in range(6):
    forward(80)
    back(30)
    left(45)
    forward(25)
    back(25)
    right(90)
    forward(25)
    back(25)
    left(45)
    back(50)
    right(60)
```

✨ Key point

One branch is 10 lines. It returns to the start, so right(60) moves to the next branch.

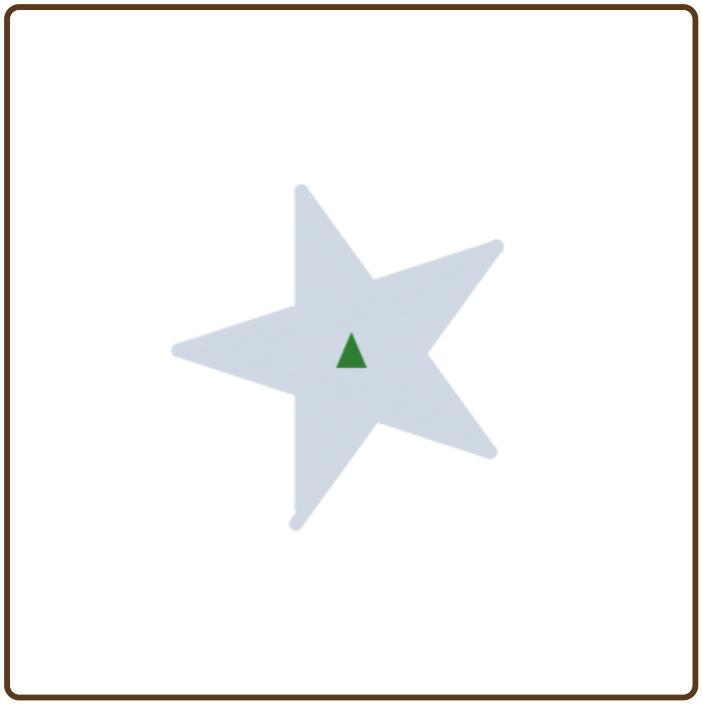
⚠ Common mistake

Forgetting the last back(50) shifts each branch.

🎬 Video: `videos/en/pydraw-08.mp4`

⑨ Star spiral ★★★★★

Variables | Math | Loops | Python syntax (colon & indent)



Model (draw over the light lines)

Same shape as block challenge 9

Problem

Use for i in range(1, 51): (i from 1 to 50), then forward(i * 4) and right(144). Change colors with pen_color('random').

🎯 **Goal** : Variable × math + turning

💡 **Hint (look if you get stuck)**

* means multiply. The line gets 4 steps longer each time.

✏ **Write your own idea**

The answer is on the next page →

📖 Answer ⑨ Star spiral ★★★★★

```
for i in range(1, 51):
    pen_color('random')
    forward(i * 4)
    right(144)
```

✨ **Key point**

The length i * 4 grows by 4 each time, and turning 144 makes a star spiral.

⚠ **Common mistake**

range(1, 50) stops after 49. Use range(1, 51) for 50 times.

🎬 Video: `videos/en/pydraw-09.mp4`

Column

How to read range

`range(6)` gives 0, 1, 2, 3, 4, 5 (six numbers). `range(10, 201, 10)` starts at 10, adds 10 each time, and stops before 201 (at 200). In Python, the last number is not included. In `360 / n`, “/” means division.

Well done!

You worked hard to the end! Let’s look back at the ideas you used in this book.

Idea	Where did it appear?
Loops	Pack “move, turn” together (① – ③)
Angles	360 ÷ corners; a star is 720 ÷ 5 (① – ③)
Nested loops	A loop inside a loop (④ ⑤)
Variables	Repeat while changing i (⑥ ⑦ ⑨)
Breaking into parts	One branch → six times (⑧)
Python	Write the same thing in words (code)

Try Free drawing to make your own patterns! Next, try “Block Maze” and “Space Star Hunt”.

PART 2

Part 2 Guide for teachers and parents

About this guide

This book covers “Block Drawing” (9 challenges) and “Python Drawing” (the same 9). Part 1 is written in simple language for students.

In Japanese elementary mathematics, drawing regular polygons with programming is an example activity in Grade 5 (“Regular polygons and circles”). Challenges ①–③ fit that lesson directly; ④ onward extends to nested loops, variables and decomposition. In junior high, the Python version can introduce text-based programming with the same shapes.

Aims

- Express properties of shapes (side length, turn angle) with ordered commands and loops.
- Reason that the turn angle is “360 ÷ number of sides” and explain it.
- Use variables to draw shapes that change step by step (spirals, etc.).

Overview of the challenges

App	#	Stage	Ideas
🐢 Block Drawing	1	① Hexagon ★	Loops, Angles
	2	② Star ★★	Loops, Angles
	3	③ Circle ★★	Loops, Angles
	4	④ Window ★★★	Nesting
	5	⑤ Flower ★★★	Nesting, Angles
	6	⑥ Square spiral ★★★★	Variables, Loops
	7	⑦ Polygons in a row ★★★★	Variables, Math, Nesting
	8	⑧ Snowflake ★★★★★	Decomposition, Loops
	9	⑨ Star spiral ★★★★★	Variables, Math
🐍 Python Drawing	1	① Hexagon ★	Loops, Angles, Python syntax (colon & indent)
	2	② Star ★★	Loops, Angles, Python syntax (colon & indent)
	3	③ Circle ★★	Loops, Angles, Python syntax (colon & indent)
	4	④ Window ★★★	Loops, Nesting, Python syntax (colon & indent)
	5	⑤ Flower ★★★	Loops, Nesting, Angles, Python syntax (colon & indent)
	6	⑥ Square spiral ★★★★	Variables, Loops, Python syntax (colon & indent)
	7	⑦ Polygons in a row ★★★★	Variables, Math, Nesting, Python syntax (colon & indent)
	8	⑧ Snowflake ★★★★★	Sequence, Decomposition, Loops, Python syntax (colon & indent)
	9	⑨ Star spiral ★★★★★	Variables, Math, Loops, Python syntax (colon & indent)

Suggested order and teaching points

Students	Challenges	Lessons	Key ideas
Grades 3–4	① – ③	1–2	Loops, angles
Grade 5 (math)	① – ③ (+ ④ ⑤)	2	Regular polygons, 360 ÷ sides
Grades 5–6	④ – ⑨	2–3	Nesting, variables, math, decomposition
Grade 6 – junior high	Python ① – ⑨	2–4	for/range, variables, syntax

Common difficulties and support

- **Inside angle vs. turn angle:** turning 60° for a triangle. → Have students walk the shape and feel “the angle that changes the direction”.
- **Times and angle don’t match:** check that times × angle = 360 (720 for the star).
- **Where to put the turn in nested loops:** confirm in words that “after each square” means outside the inner loop.
- **Variables:** for ⑥, have students write in a table how i changes (10, 20, 30…) before building.
- **Python:** post the three rules (brackets, colon, indent). Note that the end number of `range` is not included.

Example 45-minute lesson (① – ③)

Time	Activity	Support
Intro 5 min	Review regular polygons; share the goal.	Show a triangle, square and hexagon.
Main 30 min	Draw a square and triangle; predict and check the hexagon’s angle; find the rule in a table.	Predict → run → fix. Use the Worksheet 2 table.
Wrap-up 10 min	Summarize “360 ÷ sides”; write a reflection.	Compare 2–3 programs.

Assessment (example)

Area	Meets expectations (B)	Exceeds (A), e.g.
Knowledge & skills	Combines blocks (code) correctly and completes the task.	Uses loops and conditions to complete it with a short, clear program.
Thinking, judgment & expression	Finds where the error is from the result and fixes it.	Explains the rule or steps in words and makes a program that works in other cases too.
Attitude toward learning	Keeps trying even when it doesn’t work.	Shares ideas with friends and looks for better ways on their own.

Use the worksheets (1 reflection; 2, 5, 7 etc.) and the app progress (8 check card, certificate) as evidence.

Using the apps and related materials

Requirements and tips

- Works in the browser on computers, tablets and smartphones. No installation needed. An internet connection is required.
- Cleared stages are saved in that device’s browser. If devices are shared, check progress at the start of each lesson.
- **Teacher mode:** add `?sensei` to the end of the address to show an “Answer” button.
- From “⚙ Settings” you can switch Japanese/English, use the pixel font, switch to the Python version and show the certificate.

Related materials

Material	Contents
Guide (guide.html)	Goals, common mistakes, hints and answers for every task. This book is built from the same data.
Worksheets (PDF)	2 Drawing (angle table), 7 Python, 1 goal & reflection, 8 check card.
Lesson plans (Japanese)	Example 45-minute lessons (open them from each task in the Guide).
Videos (videos.html)	Short videos showing how to solve each task.
Lesson slides	01 Block Drawing, 05 Python Drawing (PowerPoint).